

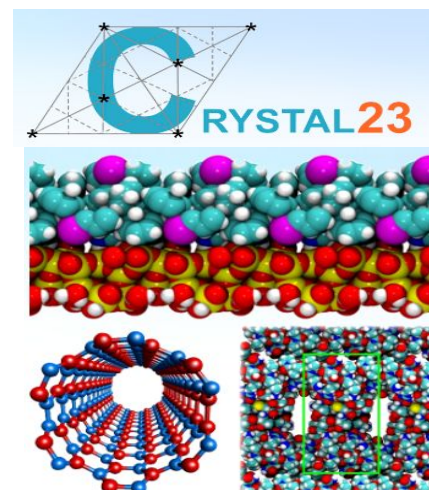
Lattice dynamics



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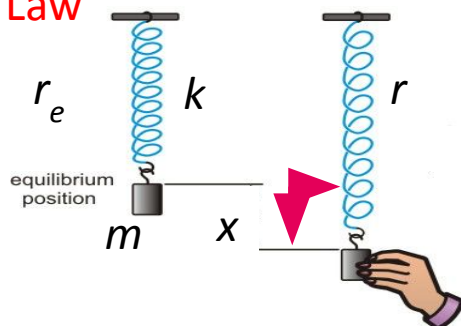
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nis CENTRE FOR
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INTERFACES
AND SURFACES

Outline

- Harmonic approximation: the molecular case
- Lattice dynamics of 1D chain
- Frequencies calculations with CRYSTAL
- LO/TO splitting
- The role of symmetry
- Data analysis
- Phonon Dispersion

The Harmonic approximation

Hooke's Law



Harmonic motion

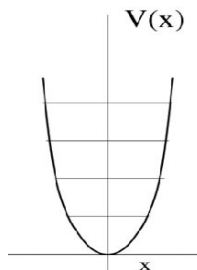
$$F = -kx$$

$$V = \frac{1}{2} kx^2$$

Newton's equation of motions

$$F = ma = -kx$$

$$v = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$



$$F = -\frac{\partial V}{\partial x}$$

$$k = \frac{\partial^2 V}{\partial x^2}$$

Molecules

Harmonic approximation

$$V(u) = V(0) + \left(\frac{\partial V}{\partial u}\right)_{u=0} \cdot u + \frac{1}{2} \left(\frac{\partial^2 V}{\partial u^2}\right)_{u=0} \cdot u^2 + \dots$$

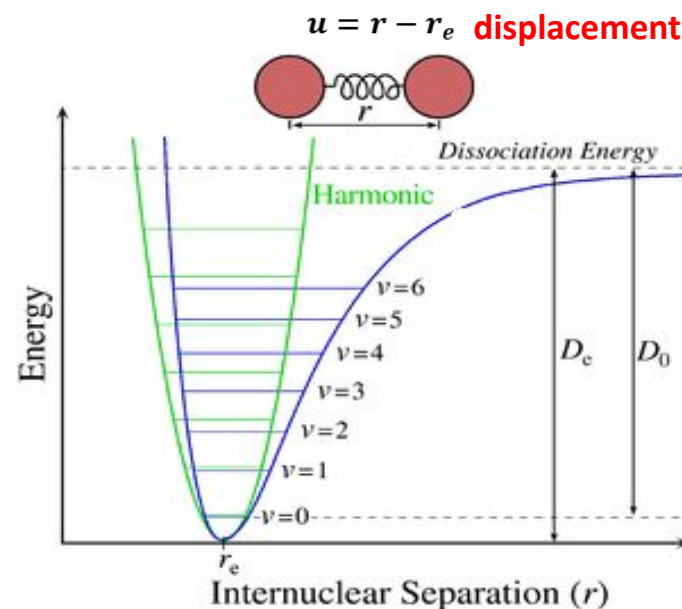
$$V(u) = \frac{1}{2} \left(\frac{\partial^2 V}{\partial u^2}\right)_{u=0} \cdot u^2 = \frac{1}{2} k u^2$$

force constant

Newton's equation of motions (classic)

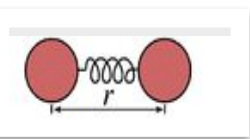
$$F = \mu a = -ku$$

$$v = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$



Nuclear Schroedindiger equation (QM)

$$H = T + V$$

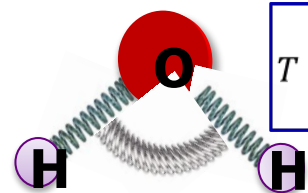
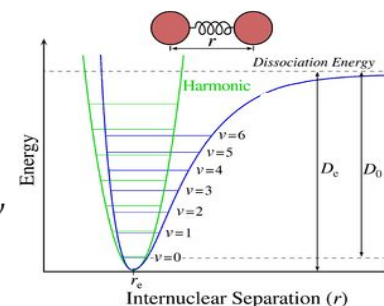


$$T = -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial u^2}$$

$$V = \frac{1}{2} \left(\frac{\partial V^2}{\partial^2 u} \right)_{u=0} = \frac{1}{2} k u^2$$

QM □ levels are quantized

$$E(n) = \left(\frac{1}{2} + n \right) h\nu$$



$$T = -\sum_{i=1}^{3N} \frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial u_i^2}$$

$$V = \frac{1}{2} \sum_{i,j}^{3N} \left(\frac{\partial V^2}{\partial u_i \partial u_j} \right)_{u_i=0, u_j=0} u_i u_j = \frac{1}{2} \sum_{i,j}^{3N} H_{ij} u_i u_j$$

3N displacements with respect to equilibrium positions

H_{ij} Hessian matrix

A system of 3N equation to be solved simultaneously
Factorization is possible in the Harmonic approximation

$$H_{ij} = \left(\frac{\partial V^2}{\partial u_i \partial u_j} \right)_{u_i=0, u_j=0} \quad \text{Hessian matrix}$$

$$D_{i,j} = \frac{H_{i,j}}{\sqrt{M_i M_j}}$$

Dynamical (mass weighted) matrix

From mass weighted coordinates
 $u_i \sqrt{M_i}$

$$\mathbf{T}^+ \mathbf{D} \mathbf{T} = \mathbf{W}$$

$\mathbf{T}$ is the matrix of the eigenfunctions
(i.e. normal coordinates $\mathbf{Q}_p$, normal modes),
 $\mathbf{W}$ is the matrix of the eigenvalues
(i.e. vibrational frequencies)

$$\nu_i = \sqrt{W_i}$$

3N independent motions (degrees of freedom)

3N-6 normal modes

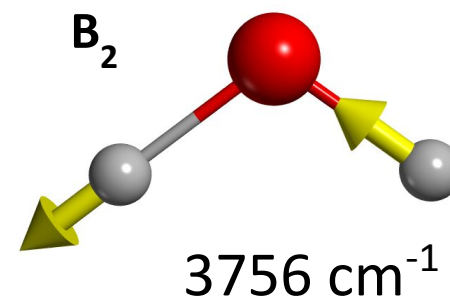
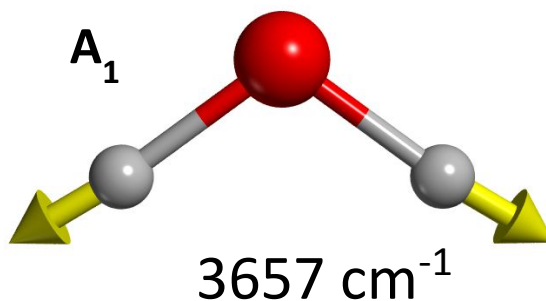
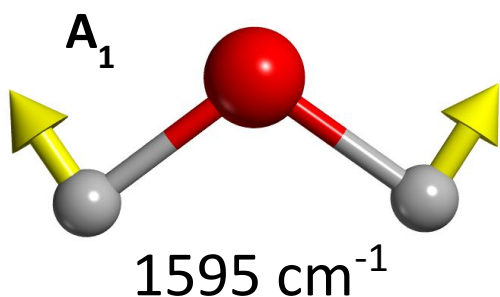
3 translations

3 rotations (2) for linear molecules

Normal modes and symmetry

Water: 3 modes IR and Raman active

$$\Gamma^{\text{vib}} = A_1 \oplus A_1 \oplus B_2$$



Methane: 9 modes

$$\Gamma^{\text{vib}} = 2F_2 \oplus E \oplus A_1$$

2 peaks

4 peaks

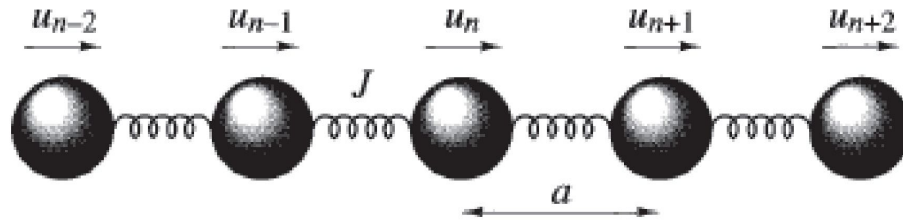
cm ⁻¹	SYM	IR	RAMAN
1693	F2	28	3
1910	E	--	104
3464	A	..	1000
3724	F2	98	928

Two active in IR, **four** in Raman

IR intensities: Km/mole

Raman intensities: arbitrary units

Lattice Vibrations: 1D chain



The simplest crystal is the one dimensional chain of **identical atoms**:

- Chain consists of a **very large number** of identical atoms with identical masses
- Atoms are separated by a distance “ a ”, the **lattice periodicity**
- Atoms move only in a direction **parallel to the chain** by a displacement u_n
- As a further approximation: only **nearest neighbours interact** (short-range forces)

Harmonic potential between couples of atoms

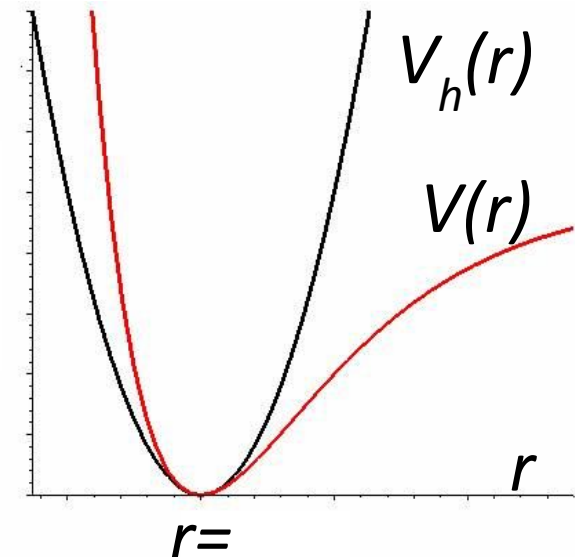
$$V_h(x) = \frac{1}{2} \left(\frac{\partial V(r)}{\partial r^2} \right)_{r=a} \cdot (r - a)^2$$

$$F = -\frac{\partial V(r)}{\partial r} = -J (r - a)$$

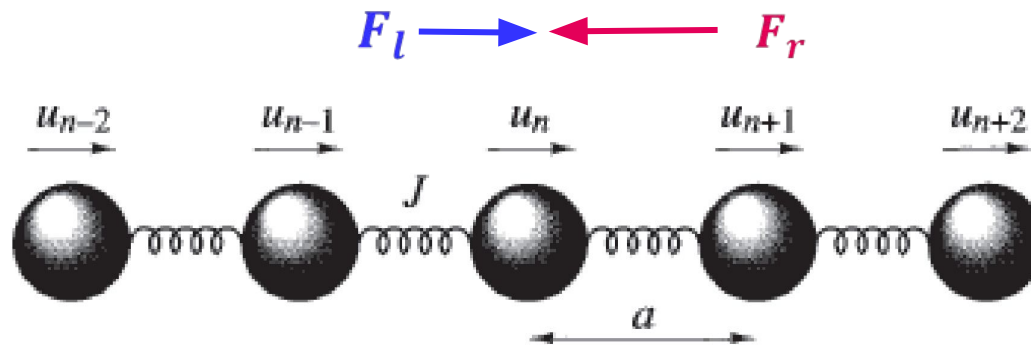
$$\mathbf{F}(\mathbf{r}) = -J \mathbf{u}$$

$\mathbf{u}$ displacement from equilibrium position

J force constant



Lattice Vibrations: 1D chain of like atoms



The Newton's equations involve the **net force** on the n^{th} atom:

Force to the right:

$$F_r = m\ddot{u} = -J(u_n - u_{n+1})$$

Force to the left:

$$F_l = m\ddot{u} = -J(u_n - u_{n-1})$$

total force:

$$F_n = F_r + F_l = -J(u_{n+1} - 2u_n + u_{n-1})$$

$$F_n = ma_n = m\ddot{u} = -J(u_{n+1} - 2u_n + u_{n-1})$$

Equation of motion of all atoms are of this form, only the value of 'n' varies

All atoms oscillate with a same amplitude $\tilde{u}$ and frequency ω . Then we can try a solution by using a travelling wave of the form:

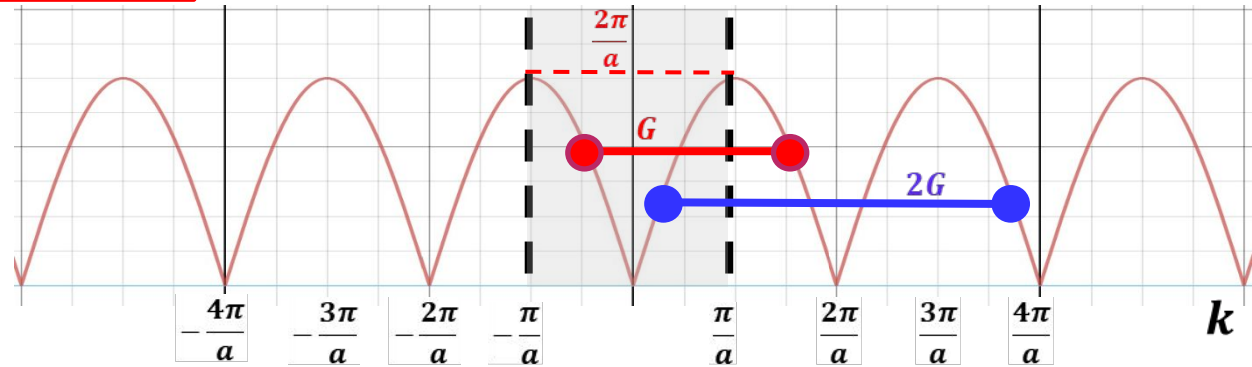
$$u_n = \tilde{u} \exp[i(kna - \omega t)]$$

$k=2\pi/\lambda$ is the wave vector describes a travelling wave of wave length $\lambda=2\pi/k$

$$\omega(k) = \left(\frac{4J}{m}\right)^{\frac{1}{2}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

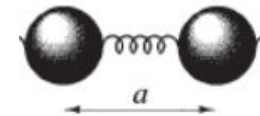
$\omega(k)$ is a periodic function with period $\frac{2\pi}{a}$

Dispersion relation for $\omega(k)$



The period $\frac{2\pi}{a}$ defines a vector G with the dimension of the reciprocal of a length (the unit cell of the monodimensional reciprocal lattice)

$$G = \frac{2\pi}{a} \text{ or } a^* = \frac{2\pi}{a}; \quad a^* a = 2\pi$$



Highlighted in grey the FBZ

Any k vector of any zone can be translated by G (or nG) in the FBZ

$$u = \tilde{u} \exp[i((k + G)na - \omega t)] = \tilde{u} \exp[i(kna - \omega t)] \exp[i(nGa)] = \tilde{u} \exp[i(kna - \omega t)] \exp[i2\pi n] = \tilde{u} \exp[i(kna - \omega t)]$$

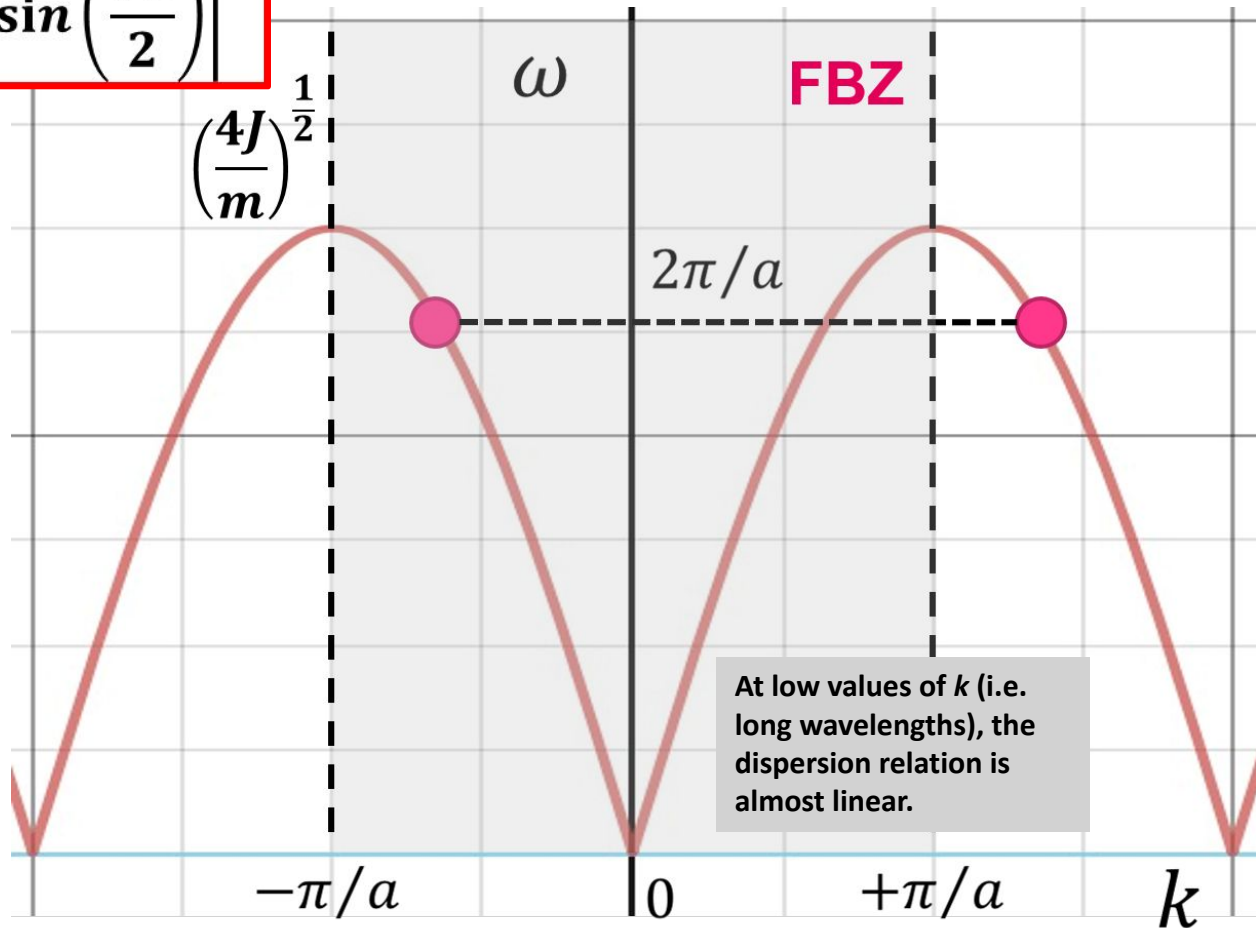
$$\omega(k) = \omega(k + G)$$

The FBZ contains all the information about the lattice

All allowed k are defined in the FBZ of the reciprocal lattice

$$\omega(k) = \left(\frac{4J}{m}\right)^{\frac{1}{2}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

$$\omega(k) = \omega(k + G)$$

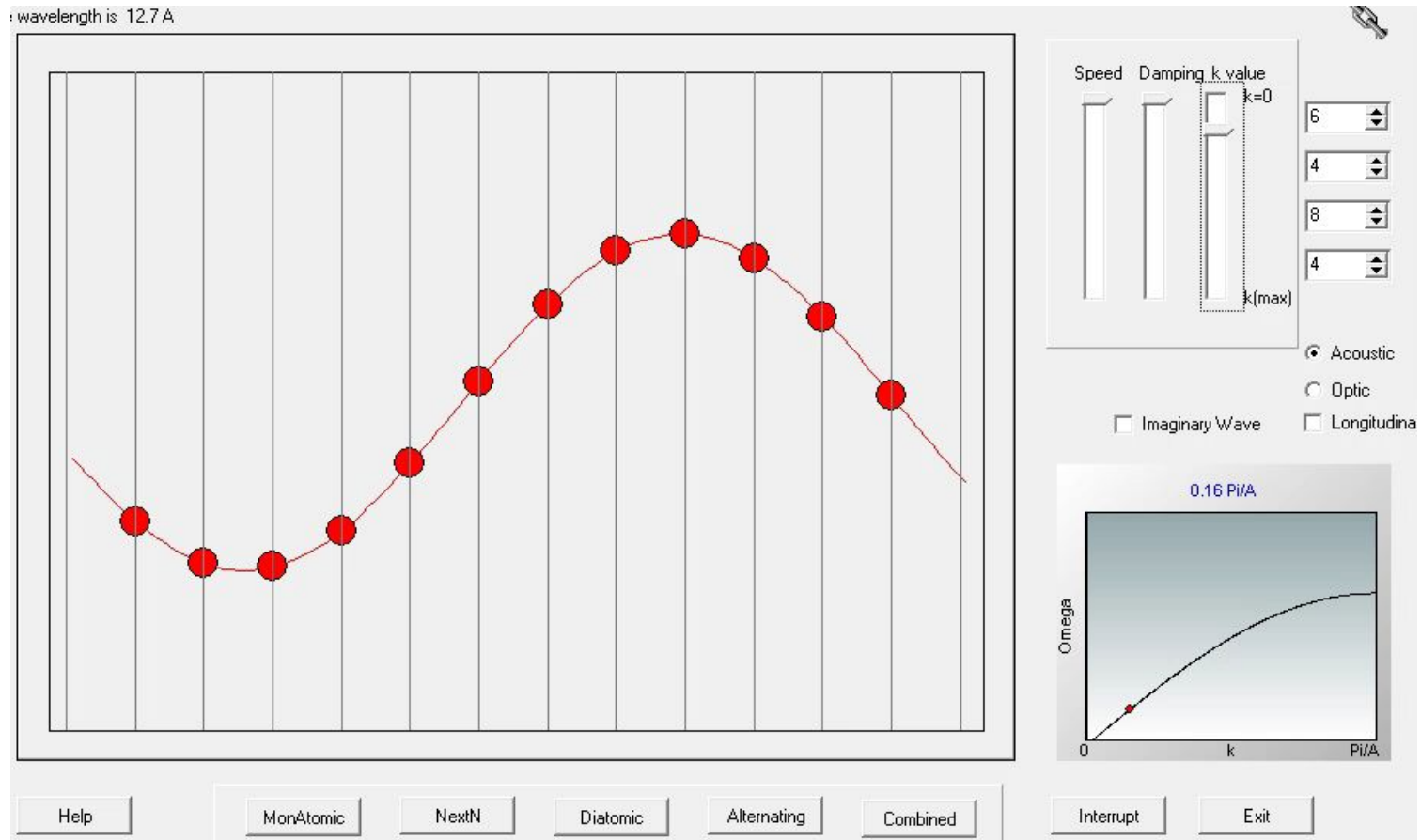


At low values of k (i.e. long wavelengths), the dispersion relation is almost linear.

At $k=0$ $\omega=0$
The normal coordinate corresponds to hindered translations

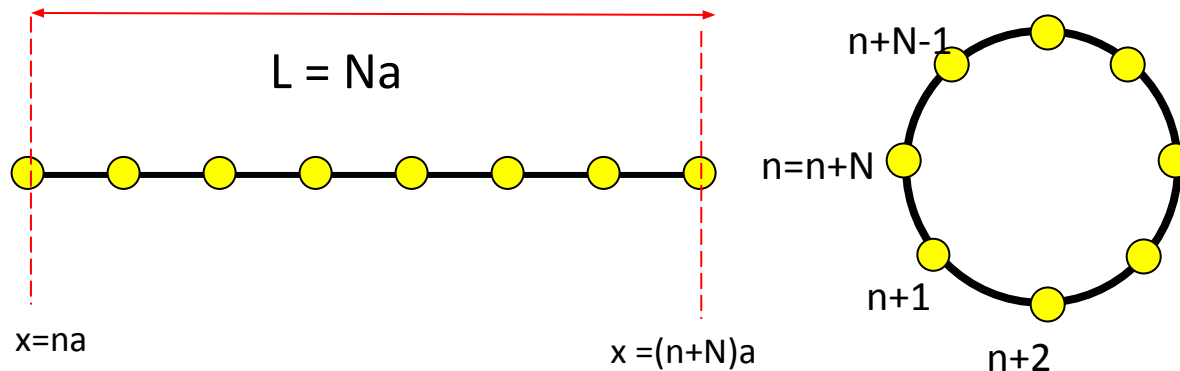
Acoustic branch

Meaning of $\omega(k)$



Counting number of $\omega(k)$

simplify problem by using periodic boundary conditions for the linear chain of atoms:



We assume atoms n and $n+N$ have the same displacement the lattice has periodic behavior, where N is very large.

Since atoms n and $n+N$ have the same displacement, we can write:

$$u_n = u_{n+N} \rightarrow ue^{i(kna-\omega t)} = ue^{i(k(n+N)a-\omega t)}$$

$$ue^{i(kna-\omega t)} = ue^{i(kna-\omega t)} e^{i(kNa)} \rightarrow 1 = e^{ikNa}$$

This sets a condition on allowed k values:

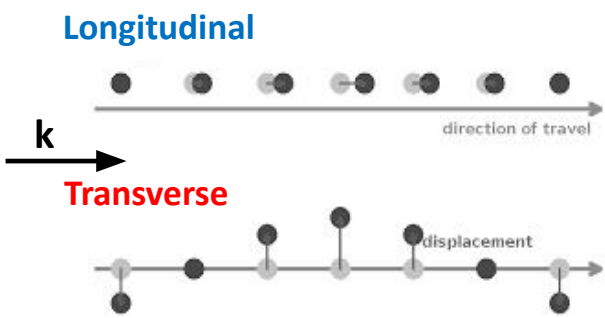
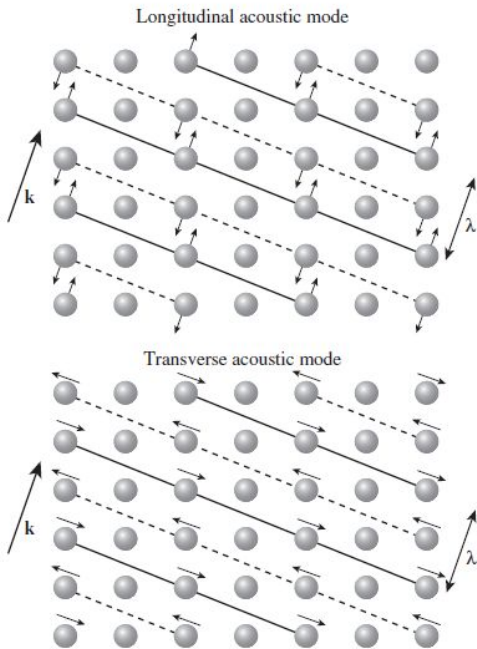
$$kNa = 2\pi m \rightarrow k = \frac{2\pi m}{Na} \quad m = 0, 1, 2, 3, \dots, N$$

But in a real system

$$N = O(10^{23})$$

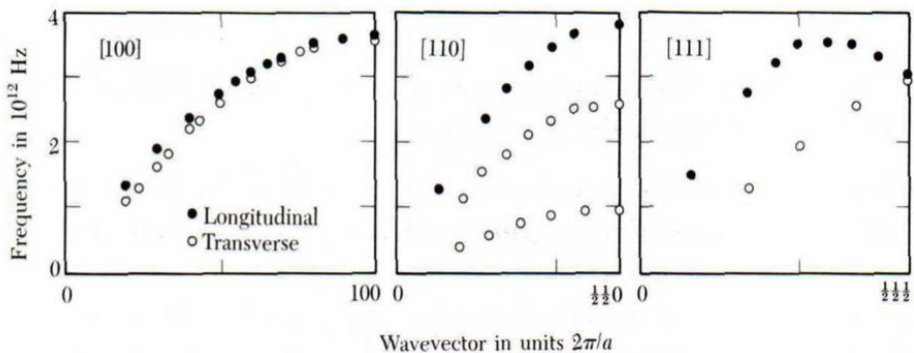
3D monatomic crystals

The one-dimensional model can be easily extended to three dimensions. We need to include motions that are perpendicular to the wave vector. The perpendicular displacements will have different force constants from the longitudinal force constants, but they will be expressed in an identical manner so that the results for the different types of motion will follow identical equations.

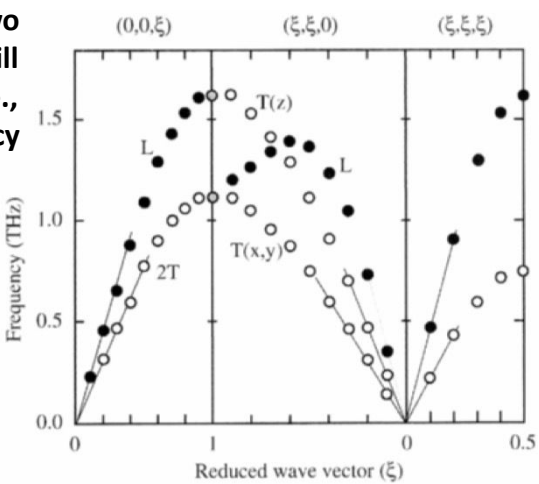


Longitudinal acoustic modes (LA) the atoms move in directions parallel to the wave vector k

Transverse acoustic modes (TA), the vibrations are in the two orthogonal directions perpendicular to the wave vector k



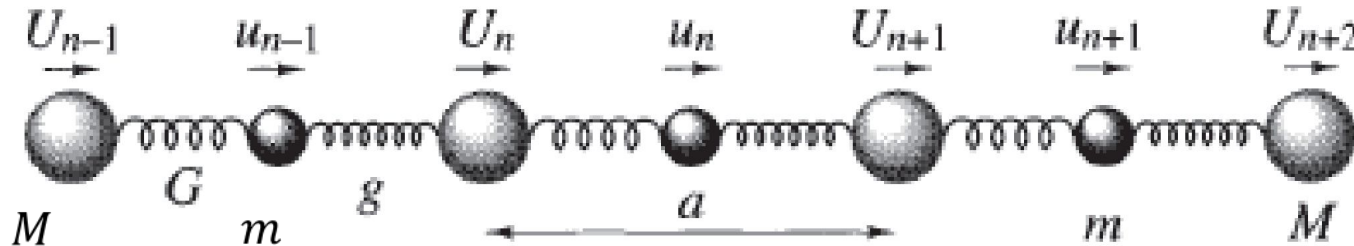
In high symmetry situations, the two transverse modes will have degenerate (i.e., equal) frequency dispersion.



Inelastic neutron scattering INS for Na (bcc)

INS for Ne (fcc)

Lattice Vibrations: 1D diatomic chain



TWO NEWTON EQUATIONS

Equation of motion for mass m (n^{th}):

$$m\ddot{u} = -G(u_n - U_n) - g(u_n - U_{n+1})$$

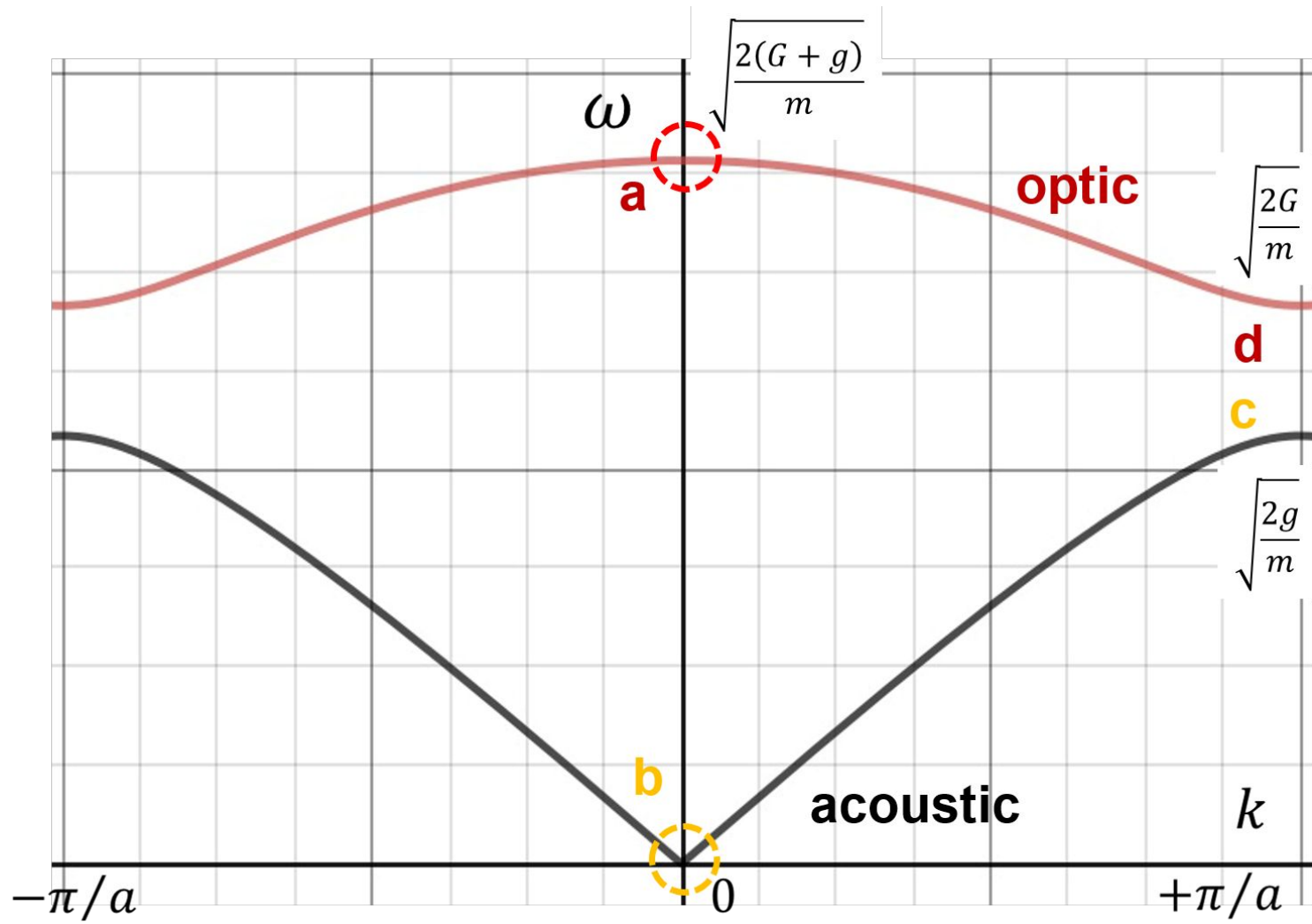
$$u_n = \tilde{u} \exp[i(kna - \omega t)]$$

Equation of motion for mass M (n^{th}):

$$M\ddot{U} = -g(U_n - u_{n-1}) - G(U_n - u_n)$$

$$U_n = \tilde{U} \exp[i(kna - \omega t)]$$

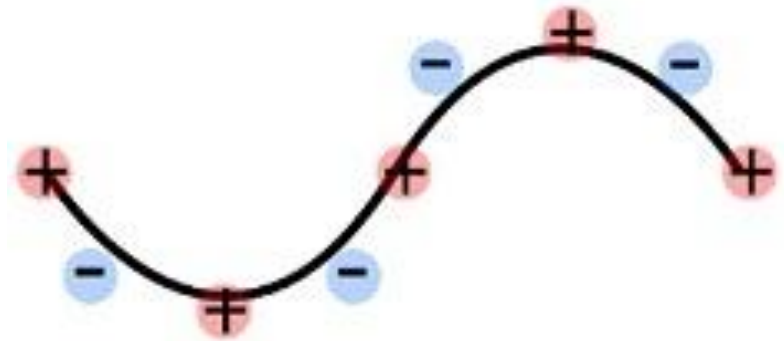
TWO NEWTON EQUATIONS: two solutions with different $\omega(k)$ dispersion curves



Atomic motion in acoustic/optic branches

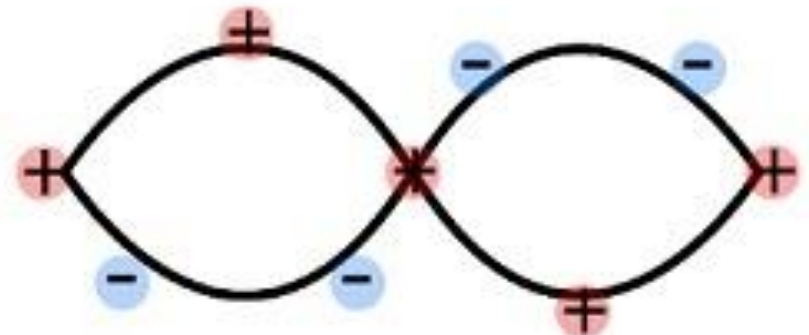
The **acoustic branch a)** has this name because it gives rise to long wavelength vibrations which travel as elastic (sound) waves and $\omega \rightarrow 0$ for $k \rightarrow 0$. The 2 atoms generally vibrate with a **small phase** difference.

Acoustical Mode

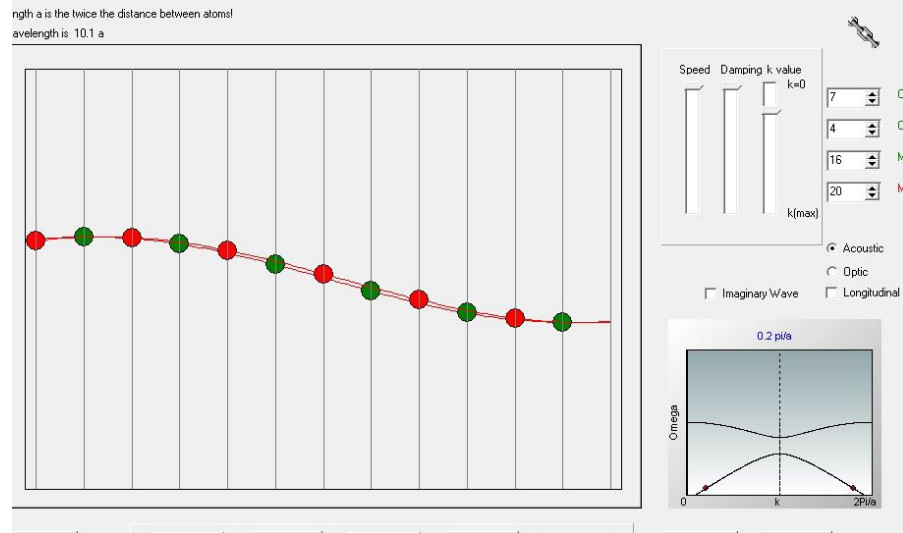


The **optic branch b)** is a higher energy vibration and they vibrate generally in **out of phase**. If the two vibrating atoms are ions, a dipole is created during the vibration that *can interact with electromagnetic radiation* giving optical properties to the crystal.

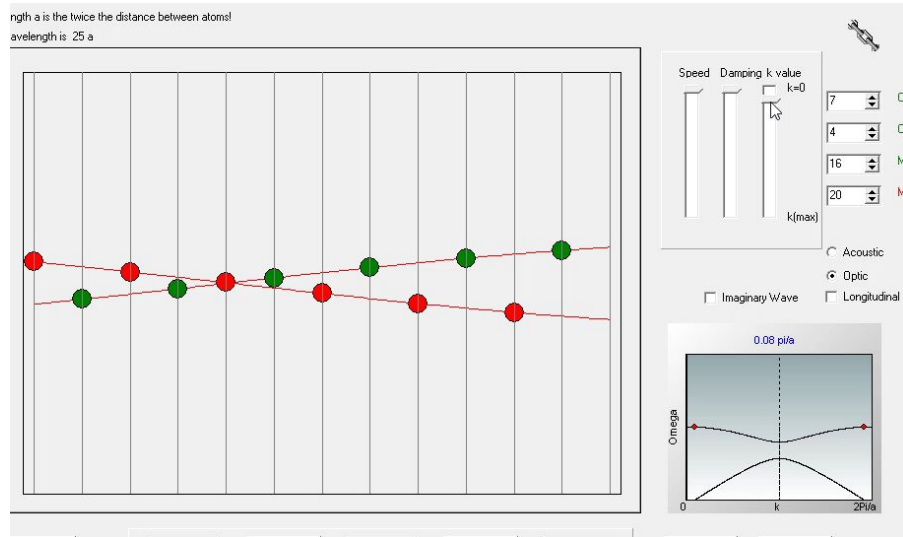
Optical Mode



Atomic motion in acoustic/optic branches



acoustic branches



optic branches

Vibrations for a 3D crystal with p-atomic base

The 3D case with 2 atoms per cell

For a 3D crystal, if each unit cell has **2** atoms, the degrees of freedom (**DOF**) are **6** for each cell

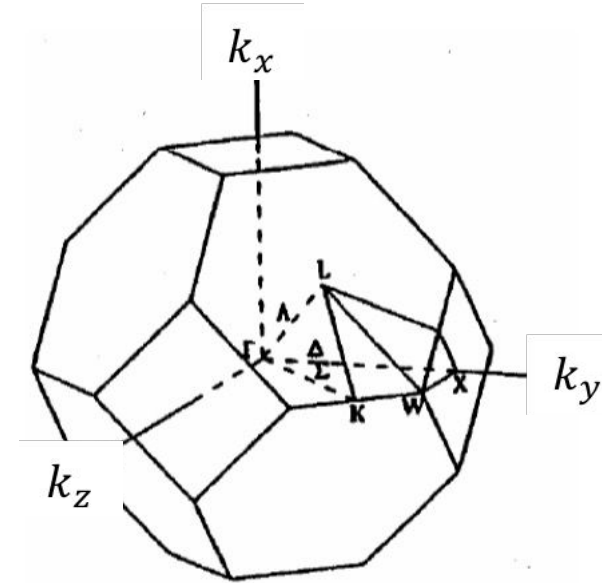
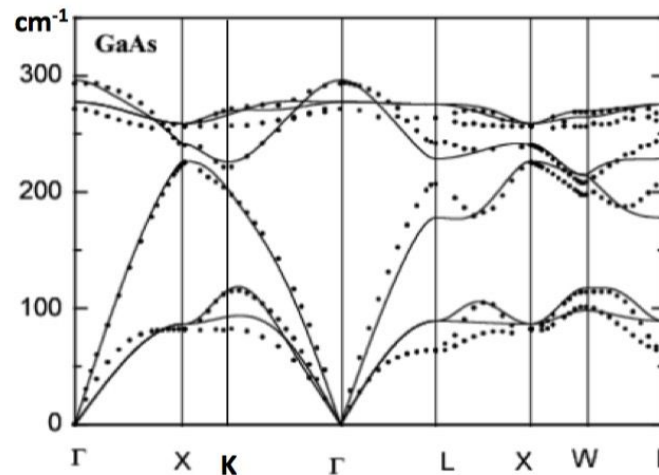
3 acoustic branches, (**L**, **T1** e **T2**)

3 optical branches, (**L**, **T1** e **T2**)

2 atoms (Ga,As)

3 **acoustic** branches L, T1 e T2

3 **optical** branches L, T1 e T2



The 3D case with p atoms per cell

3p branches for each cell (**L**, **T1** e **T2**)

3 acoustic branches, (**L**, **T1** e **T2**)

3(p-1) **optical** branches. (**L**, **T1** e **T2**)

N unit cells, then each branch has **N** normal modes

As a result, the total number of normal modes of the whole crystal is **3pN** (= total DOF of this crystal).

Phonons

Moving to quantum mechanic results are similarbut.....
the energy levels of the harmonic oscillator are **quantized**.

The **quantum of vibration is called a phonon** in analogy with the photon, which is the quantum of the electromagnetic wave.

We know that the allowed energy levels of the harmonic oscillator are given by:

$$E = (n + \frac{1}{2}) \hbar \omega$$

In a system with s atom per cell the energy of an oscillator is defined by the branch p , $p=1, \dots, s$, and by the wavevector k , $k=0, \dots, N$

$$E_{kp} = (n_{kp} + \frac{1}{2}) \hbar \omega_p(\mathbf{k})$$

The total vibrational energy of the crystal is the sum of the energies of the individual modes, so that:

$$E = \sum_{kp} E_{kp} = \sum_{kp} (n_{kp} + \frac{1}{2}) \hbar \omega_p(\mathbf{k})$$

Phonons: matter-radiation interaction

Phonons can interact with radiation and matter

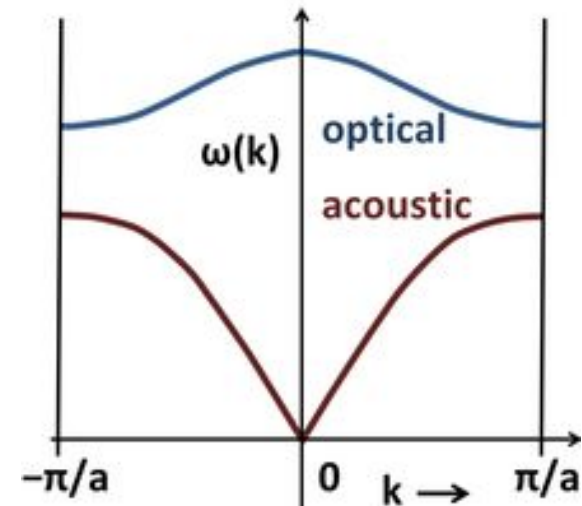
- **Phonon - Photon interaction**

- **Optic** modes at $\mathbf{k} \rightarrow 0$

- Absorption: **IR**
 - Scattering: **Raman**

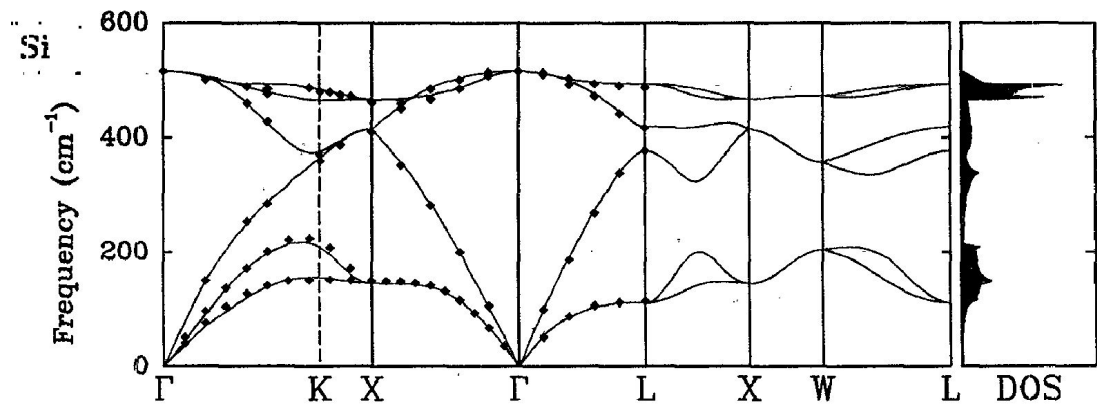
- **Acoustic** modes at $\mathbf{k} \rightarrow 0$

- Scattering: Brillouin (elastic constants)



- **Phonon-Neutron interaction**

- Inelastic Neutron Scattering (INS) (phonon dispersion)



Dynamical Matrix

In lattice dynamics the central role is played by the *Dynamical Matrix*

$$D_{ij}(\mathbf{k}) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{G}} \left[\frac{\partial^2 E}{\partial \mathbf{u}_i^0 \partial \mathbf{u}_j^{\mathbf{G}}} \right]_0 \exp[-i\mathbf{k} \cdot \mathbf{R}(\mathbf{G})]$$

$\omega(k)$ dispersion comparable with INS

harmonic approximation of the potential energy (second derivatives)

Where:

M_i is the mass of the atom associated to the i -th coordinate;
 u_i is the cartesian atomic displacements of the i -th coordinate;
 $\mathbf{R}(\mathbf{G}) = \mathbf{x}_i(0) - \mathbf{x}_j(\mathbf{G})$ ($\mathbf{G}$ as a translation vector)

From diagonalization of the dynamical matrices $\mathbf{D}$ one gets the phonon frequencies at each $\mathbf{k}$ point

$$\mathbf{T}(\mathbf{k})^+ \mathbf{D}(\mathbf{k}) \mathbf{T}(\mathbf{k}) = \mathbf{V}(\mathbf{k})$$

$\mathbf{T}(\mathbf{k})$ is the matrix of the eigenfunctions (i.e. normal coordinates Q_p , normal modes),
 $\mathbf{V}(\mathbf{k})$ is the matrix of the eigenvalues (i.e. vibrational frequencies)

$$\omega_i = \sqrt{V_i}$$

with the condition

$$\mathbf{T}(\mathbf{k})^+ \mathbf{T}(\mathbf{k}) = \mathbf{1}$$

Frequencies Calculation in CRYSTAL

CRYSTAL computes vibrational frequencies at Γ point ($\mathbf{k}=0$)

$$D_{ij}(\mathbf{k} = 0) = \frac{1}{\sqrt{M_i M_j}} \left[\frac{\partial^2 E}{\partial u_i^0 \partial u_j^0} \right]_0$$

The second-derivatives matrix is computed by *numerical differentiation of the analytical first-derivatives (gradients)*

$$\left[\frac{\partial^2 E}{\partial u_i^0 \partial u_j^0} \right]_0 = \left[\frac{\partial g_{i,0}}{\partial u_j^0} \right]_0 = \frac{g_{i,0}(0, \dots, \delta u_j^0, \dots, 0) - g_{i,0}(0, \dots, -\delta u_j^0, \dots, 0)}{2\delta u_j^0}$$

Special properties of the symbol Γ point ($\mathbf{k}=0$):

- $D(0)$ is simple to calculate
- Three modes have zero frequency (*acoustic branch – “translations”*)
- $D(0)$ possesses the point symmetry of the crystal
- Isotopic shift can be calculated at no cost!
 - Γ point modes give rise to IR and Raman spectra

Frequencies Calculation in CRYSTAL

CRYSTAL computes vibrational frequencies at Γ point ($\mathbf{k}=0$)

$$D_{ij}(\mathbf{k}=0) = \frac{1}{\sqrt{M_i M_j}} \left[\frac{\partial^2 E}{\partial u_i^0 \partial u_j^0} \right]_0$$

$\mathbf{T}$ (eigenvector matrix of the $\mathbf{D}$ matrix) transforms the cartesian atomic coordinates $u_{\alpha j}$ into the p -th normal coordinates $\mathbf{Q}_p$ and the Born charge tensors in the basis of the vibrational modes.

$$\mathbf{Q}_p = \sum_{\alpha i} \left(\frac{t_{p,\alpha i}}{\sqrt{M_\alpha}} \right) u_{\alpha i}$$

$$\bar{\mathbf{Z}}_{p,i} = \sum_{\alpha j} \frac{t_{p,\alpha j} \mathbf{Z}_{\alpha,ij}^*}{\sqrt{M_\alpha}}$$

Born tensors in the basis of the vibrational modes

Born tensors in the cartesian basis

$$\mathbf{Z}_{\alpha,ij}^* = \frac{\partial}{\partial u_{\alpha j}} \boldsymbol{\mu}_i$$

$\boldsymbol{\mu}$ =cell dipole moment (polarization per unit cell)

IR

$$A_p \propto d_p \bar{\mathbf{Z}}_{p,j}^2$$

The intensity

Raman

$$I_{ij}^k \propto C(\alpha_{ij}^k)^2 \quad \alpha_{ij}^k = \frac{\partial^3 E^{\text{TOT}}}{\partial Q_k \partial \mathcal{E}_i \partial \mathcal{E}_j}$$

- BERRY's PHASE (**INTPOL** default)
- LOCALIZED WANNIER FUNCTIONS (**INTLOC**)
- CPHF/KS (**INTCPHF**)

- CPHF/KS (**INTCPHF**)
also IR intensities are computed

CRYSTAL IR and Raman SPECTRA

```
piropo  
CRYSTAL
```

```
.....
```

```
FREQCALC
```

```
INTENS
```

```
IRSPEC
```

```
END
```

```
END
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```
.....
```

Piropo.d12

```
FREQCALC
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INTENS
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INTRAMAN
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```
INTCPHF
```

```
ENDCPHF
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```
RAMSPEC
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```
END
```

```
END
```

```
runcry17 Piropo &
```

```
□ Piropo.out
```

```
□ Piropo.freqinfo (FREQINFO.DAT)
```

```
□ Piropo.irspec (IRSPEC.DAT)
```

**Unit for
RESTART**

```
runcry17 Piropo &
```

```
□ Piropo.out
```

```
□ Piropo.freqinfo (FREQINFO.DAT)
```

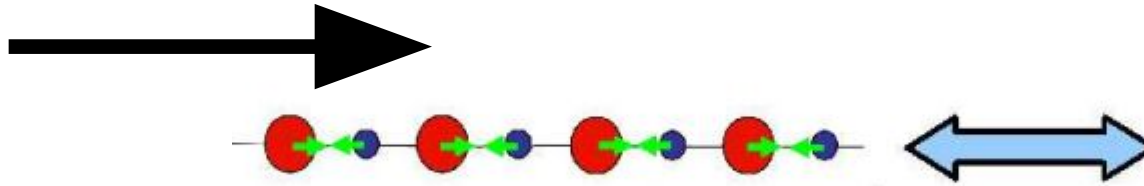
```
□ Piropo.ramanpec (RAMANSPEC.DAT)
```

**Units to plot
the spectra**

Frequencies Calculation in CRYSTAL - II

In polar crystals, long range Coulomb effects give rise to macroscopic electric fields for longitudinal optic modes (LO) at $\mathbf{k} \rightarrow 0$ adding additional restoring force

□ the LO modes are upshifted (*LO-TO splitting*):



LO-TO splitting is computed by including a non-analytical term which **depends on the electronic dielectric tensor ϵ_∞** and on the *Born effective charge tensor* associated to each atom.

$$D_{ij}(\mathbf{k} \rightarrow 0) = D_{ij}^{an}(\mathbf{k} = 0) + D_{ij}^{na}(\mathbf{k} \rightarrow 0)$$

$$D_{ij}^{na}(\mathbf{k} \approx 0) = \frac{4\pi}{V} \frac{(\mathbf{k} \cdot \mathbf{Z}^*)_i (\mathbf{k} \cdot \mathbf{Z}^*)_j}{\mathbf{k} \cdot \mathbf{k}}$$

Where:

V is the volume of the unit cell;

$\mathbf{Z}^*$ is the Born effective-charge tensor

ϵ_∞ is the electronic dielectric-constant tensor (CPHF/KS)

All those quantities can be computed by CRYSTAL

Frequencies Calculation in CRYSTAL - II

Longitudinal Optical (LO) modes

$$\varepsilon_{ij}^0 = \varepsilon_{ij}^\infty + \frac{4\pi}{\Omega} \sum_p \frac{\bar{Z}_{p,i} \bar{Z}_{p,j}}{\omega_p^2}$$

Static dielectric constant Ionic contribution

$$\varepsilon_{ij}(\omega) = \varepsilon_{ij}^\infty + \frac{4\pi}{\Omega} \sum_p \frac{\bar{Z}_{p,i} \bar{Z}_{p,j}}{\omega_p^2 - \omega^2 - i\omega\gamma_p}$$

The dielectric function is obtained with the Drude-Lorentz classical dispersion relation (damped harmonic oscillator model)

$$R(\omega) = \left| \frac{\sqrt{\varepsilon(\omega)} - 1}{\sqrt{\varepsilon(\omega)} + 1} \right|^2$$

Reflectance spectra

$\varepsilon^0 \rightarrow$ static (low-frequency) dielectric constant

$\varepsilon^\infty \rightarrow$ electronic (high-frequency) dielectric constant

$\omega_p \rightarrow$ p-th frequency eigenvalue

$\Omega \rightarrow$ unit cell volume

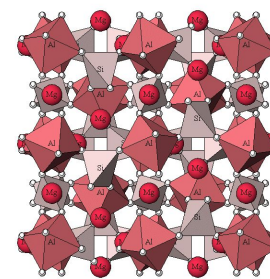
```

title
CRYSTAL
.....
FREQCALC
INTENS
DIELTENS
2.4986 0. 0.       $\varepsilon_\infty$  the electronic
0. 2.4986 0.      dielectric-constant
0.0 0.0 1.9682    tensor (CPHF/KS) or exp
IRSPEC
DIelfun           units for CRYSPLOT
IRREFR
END
ENDFREQ
END
    
```

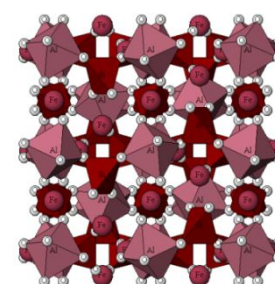
The symmetry: the GARNET FAMILY



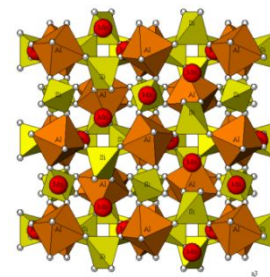
X	Y	Name
Mg	Al	Pyrope
Ca	Al	Grossular
Fe	Al	Almandine
Mn	Al	Spessartine
Ca	Fe	Andradite
Ca	Cr	Uvarovite



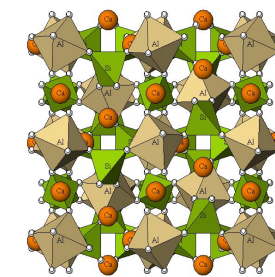
Pyrope



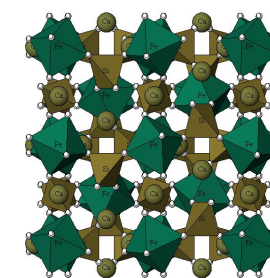
Almandine



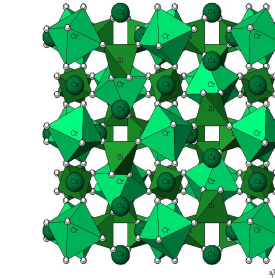
Spessartine



Grossular



Andradite



Uvarovite

Cubic **Ia-3d**

160 atoms in the UC (80 in the primitive)

$$\Gamma^{\text{rid}} = 3A_{1g} + 5A_{2g} + 8E_g + 14F_{1g} + 14F_{2g} + 5A_{1u} + 5A_{2u} + 10E_u + 18F_{1u} + 16F_{2u}$$

17 IR (F_{1u}) and 25 RAMAN (A_{1g} , E_g , F_{2g}) active modes

Hessian construction and symmetry

Each SCF+Gradient calculation provides one line of H_{ik}
80 atoms = 240+1 SCF+G calculations with low (null) symmetry

1. Point symmetry is used to generate lines of atoms symmetry related
2. Other symmetries (among x, y, z lines; translational invariance) further reduce the number of required lines

At the end only 9 out of 241 SCF+G calculations are required

DATA ANALYSIS: MODE CLASSIFICATION

Assignment of symmetry of the mode

IR/Raman activities

automatically performed by the code

MODES		EIGV (HARTREE*2)	FREQUENCIES		IRREP	IR	INTENS (KM/MOL)	RAMAN
			(CM ⁻¹)	(THZ)				
1-	3	0.3364E-17	0.0000	0.0000	(Flu)	A (	0.00)	I
4-	6	0.1930E-06	96.4150	2.8904	(F2g)	I (	0.00)	A
7-	9	0.2365E-06	106.7351	3.1998	(F2u)	I (	0.00)	I
10-	12	0.2681E-06	113.6400	3.4068	(Flu)	A (	3100.69)	I
13-	15	0.3900E-06	137.0582	4.1089	(Flu)	A (	31.78)	I
16-	16	0.4224E-06	142.6363	4.2761	(A2u)	I (	0.00)	I
17-	19	0.4571E-06	148.3824	4.4484	(Flg)	I (	0.00)	I
20-	22	0.5666E-06	165.1989	4.9525	(Flg)	I (	0.00)	I
23-	25	0.5957E-06	169.3892	5.0782	(F2g)	I (	0.00)	A
26-	28	0.7099E-06	184.9183	5.5437	(Flu)	A (	3303.16)	I
29-	31	0.7983E-06	196.0994	5.8789	(Flg)	I (	0.00)	I
32-	34	0.8571E-06	203.1900	6.0915	(F2g)	I (	0.00)	A
35-	37	0.8902E-06	207.0793	6.2081	(F2u)	I (	0.00)	I
38-	39	0.9064E-06	208.9561	6.2643	(Eg)	I (	0.00)	A
40-	41	0.9186E-06	210.3539	6.3063	(Eu)	I (	0.00)	I
42-	44	0.9419E-06	212.9980	6.3855	(Flu)	A (	0.01)	I
45-	47	0.1009E-05	220.4586	6.6092	(F2u)	I (	0.00)	I
48-	48	0.1166E-05	237.0220	7.1057	(A2g)	I (	0.00)	I
49-	51	0.1389E-05	258.6288	7.7535	(Flu)	A (	645.55)	I
.....								
233-	235	0.1961E-04	971.9773	29.1391	(Flg)	I (	0.00)	I
236-	236	0.2203E-04	1030.2214	30.8853	(A2g)	I (	0.00)	I
237-	237	0.2278E-04	1047.4163	31.4008	(Alu)	I (	0.00)	I
238-	240	0.2368E-04	1068.0768	32.0201	(F2g)	I (	0.00)	A

DATA ANALYSIS: CRYSPLOT

CRYSPLOT

Visualizing Crystal Properties

<http://crysplot.crystalsolutions.eu/index.html>

G. Beata, G. Perego, B. Civalleri



CRYSPLOT

Make a plot ▾

What is

Sample files

Documentation ▾

Contacts

Band structure

Band structure plot

Density of states

Density of states plot

Crystal orbital overlap population

Crystal orbital Hamiltonian population

Unified plot of band structure and density of states

Vibrational spectra & animations

Electron charge density and

Electrostatic potential

Electrostatic potential map

Electron charge density map

Difference map from a single file

Difference map from multiple files

Electron charge density profile

Electron momentum density

Compton profiles

Autocorrelation function

Elastic properties

Vibrational spectra

Infrared spectra

Infrared spectra comparison

Raman spectra

Raman spectra comparison

Raman and infrared spectra

Reflectance spectra

Complex dielectric function

Phonon dispersion

Phonon band structure

Phonon density of states

Volumetric data

Transport properties

Electron conductivity

Seebeck coefficient

Electron thermal conductivity

Seebeck $^2\sigma$

TOPOND

Topological analysis map

Topological analysis difference map

Geometry optimization

Geometry structure

Pair correlation function



Environment for plotting properties of the CRYSTAL code.

What is CRYSPLOT?

CRYSPLOT is an online web-oriented tool to visualize computed properties of periodic systems. It is targeted for plotting properties of crystalline solids computed with CRYSTAL code and in particular, one can represent band structure and density of states, among many others.

Why use CRYSPLOT?

CRYSPLOT is a modern and flexible online tool that make CRYSTAL more users friendly. CRYSPLOT is also totally free and easy to use. It permits the modification and customization of plots to meet the standards required for scientific graphics.

How CRYSPLOT is made?

CRYSPLOT has been designed with advanced and freely available graphical javascript libraries, namely: Plotly. CRYSPLOT allows users to read information from CRYSTAL output files and automatically generate charts for a given property.

DATA ANALYSIS: Isotopic substitution

$$D_{ij}(\mathbf{k} = 0) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{G}} \left[\frac{\partial^2 E}{\partial u_i^0 \partial u_j^{\mathbf{G}}} \right]_0$$

Changing the mass of one atomic species at a time

The dynamic matrix is re-computed at no cost by changing M_i

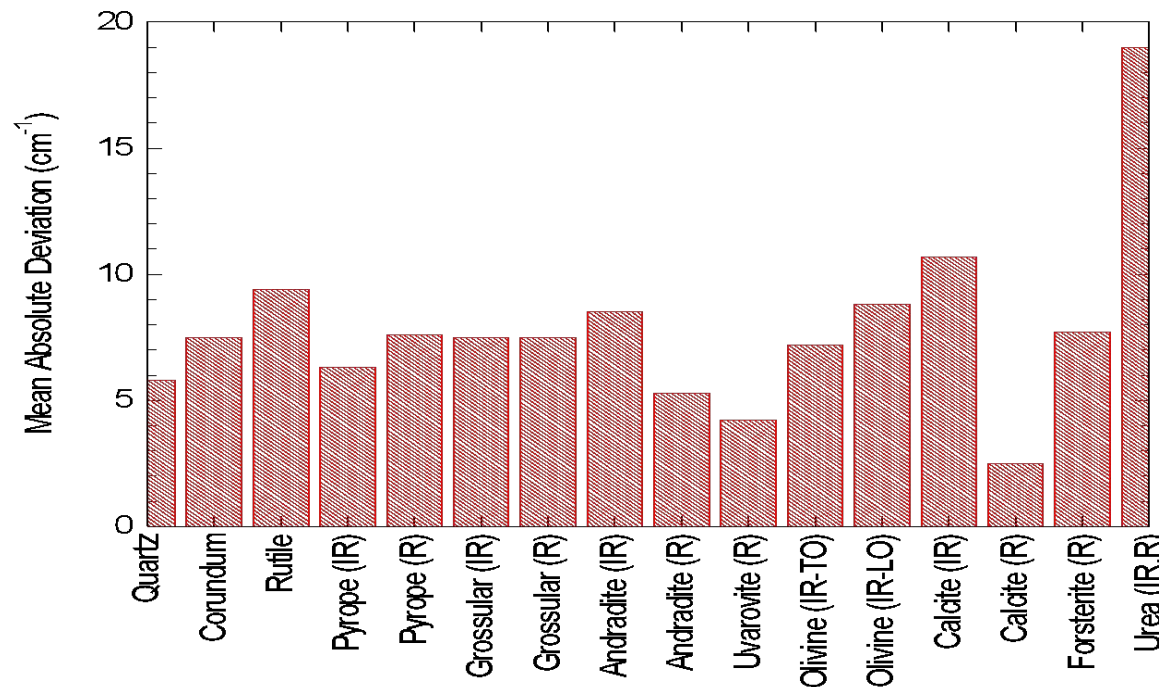
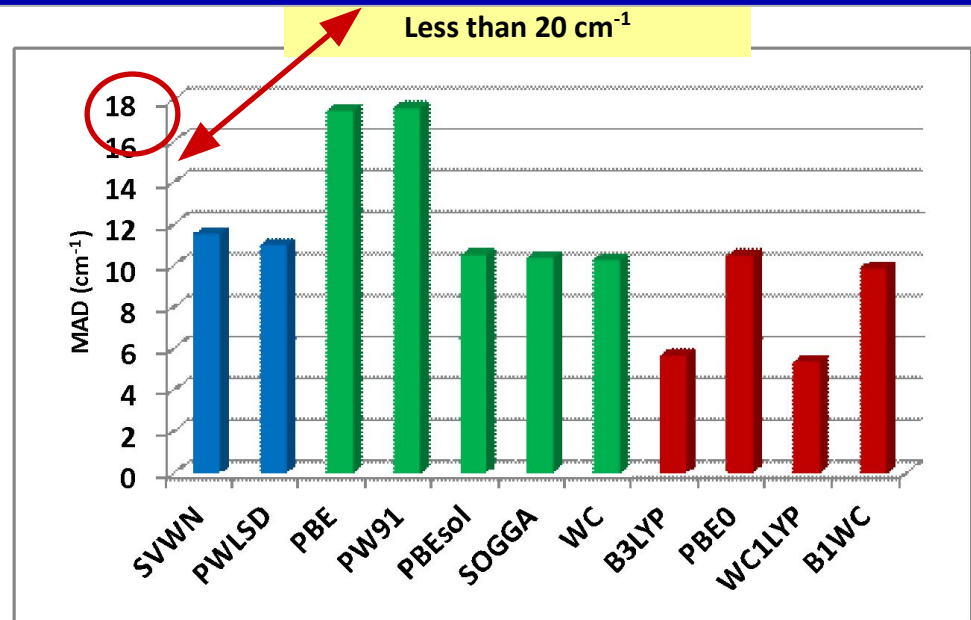
As a tool for the assignment of the modes and for the interpretation of the spectrum

- Natural isotope masses (e.g. ^{29}Al for ^{27}Al) (experimental data available for comparison)
- Any value of mass variations

```
.....  
FREQCALC  
INTENS  
ISOTOPES  
3  
1   new mass  
2   new mass  
3   new mass  
RESTART  
END  
END  
.....
```

Accuracy: DFT functionals vs experiment

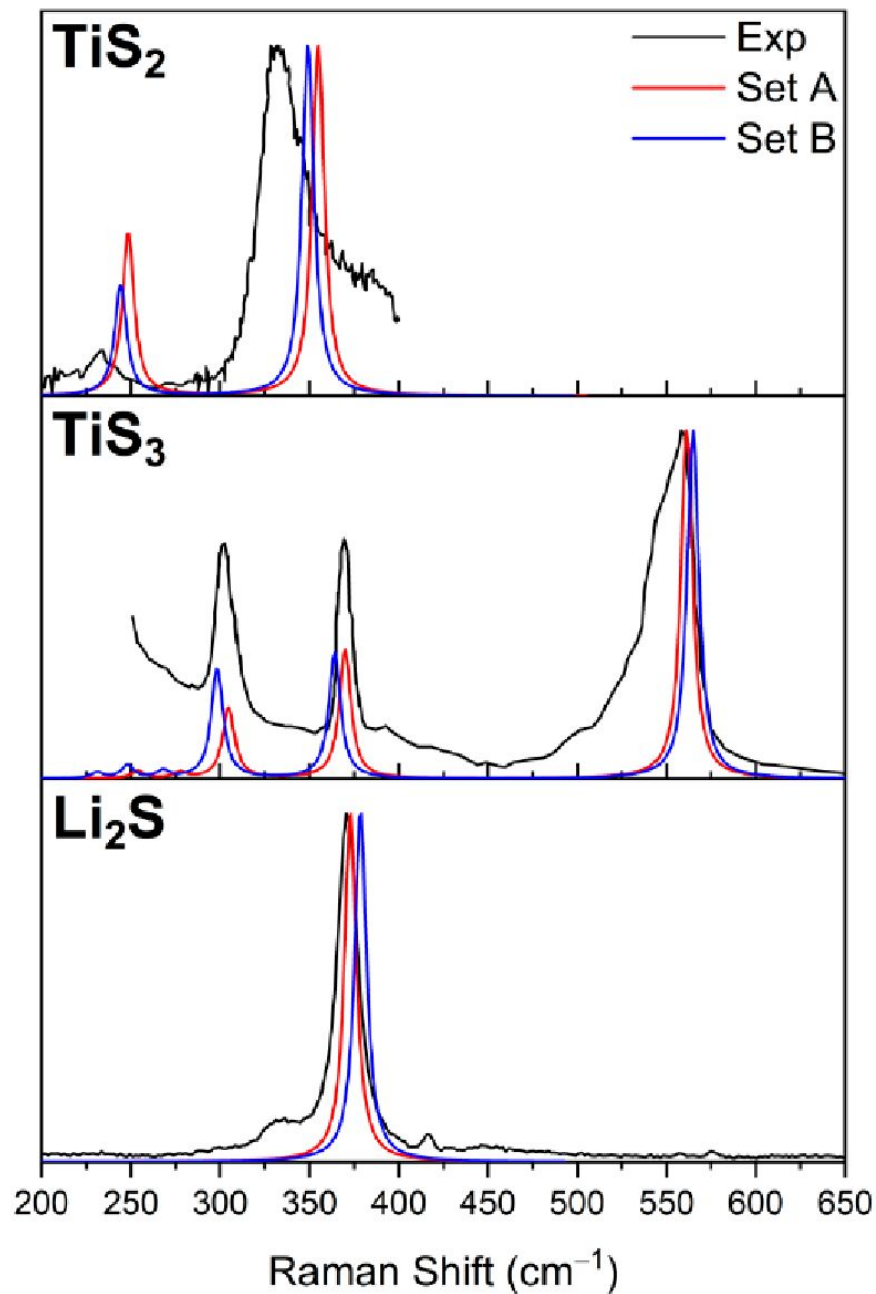
- **Four different systems:** pyrope, forsterite, quartz and alumina
- Dataset of **134** vibrational frequencies (IR and Raman data)
- **11** DFT methods: LDA, GGA (standard and for solids), **hybrids**
- **Hybrid** methods, in particular, B3LYP and WC1LYP **give the lowest** MeanAbsoluteDeviation



Accuracy: B3LYP
vs experiment

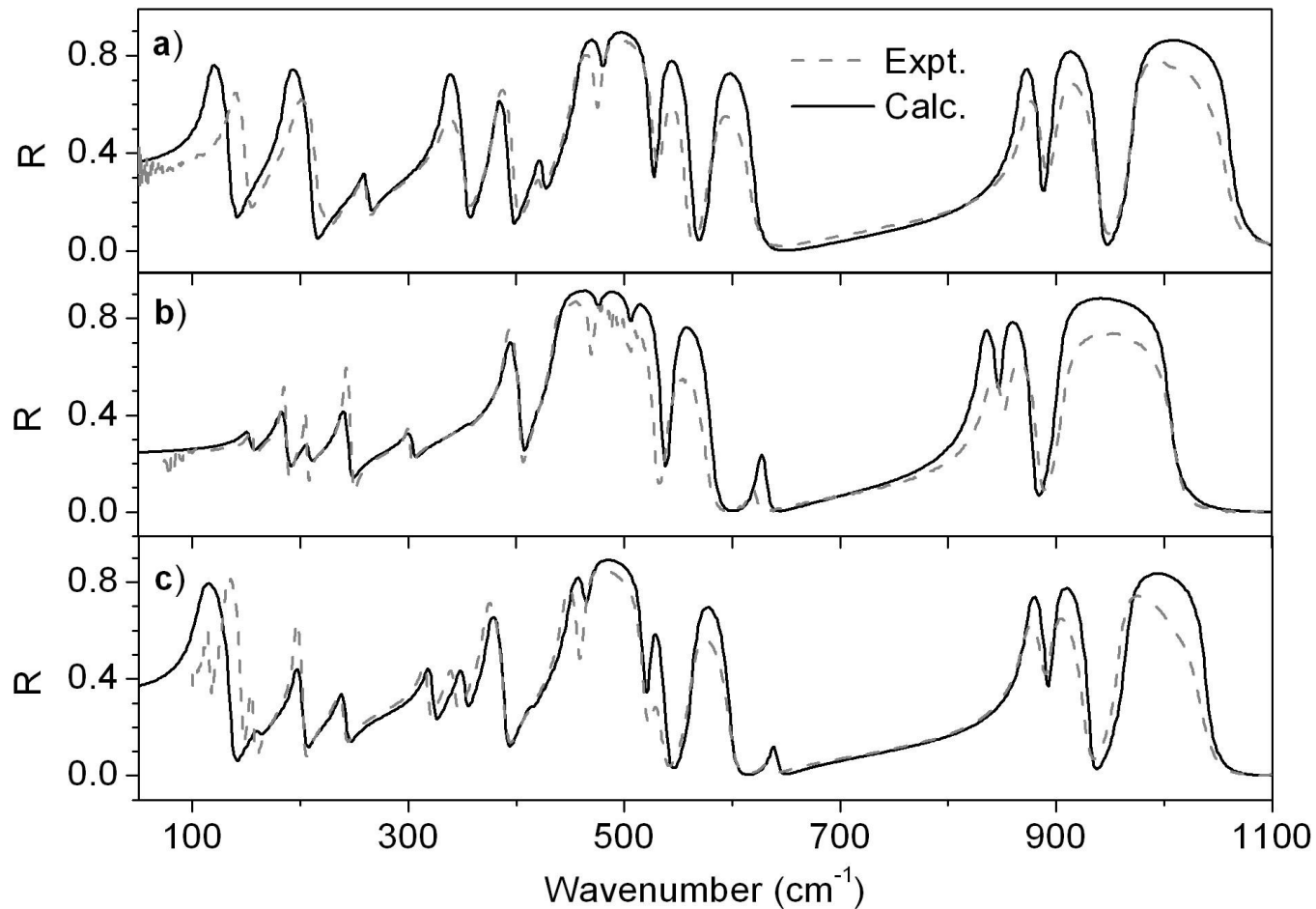
IR=Infrared
R=Raman

Demichelis, Civalleri, Ferrabone, Dovesi,
IJCQ (2010)



Rocca, R. et al. Disordered Rock-Salt Type Li_2TiS_3 as Novel Cathode for LIBs: A Computational Point of View. *Nanomaterials* **2022**, 12, 1832.

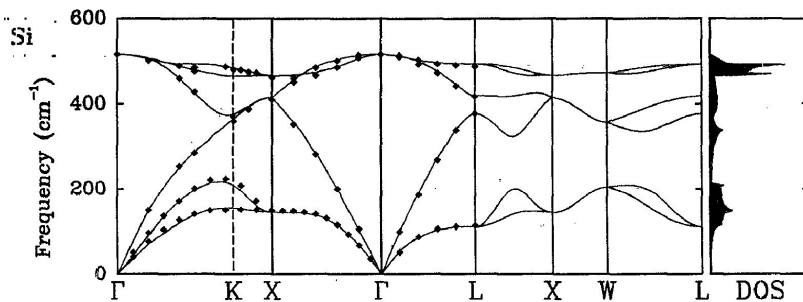
IR reflectance spectrum



Comparison of computed and experimental IR reflectance spectra for garnets: a) pyrope b) grossular c) almandine .

Phonon Dispersion

Frequencies are calculated in Γ .



To calculate phonon bands and DOSS, INS spectra

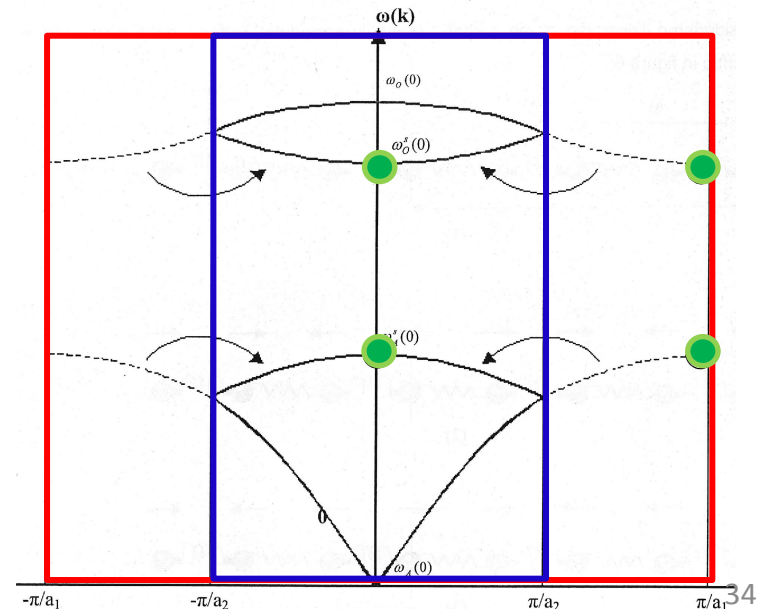
$$Q_{vib}(T) = \prod_{s,k} \frac{e^{-\frac{\hbar\omega(k,s)}{2k_B T}}}{1 - e^{-\frac{\hbar\omega(k,s)}{k_B T}}}$$

To calculate partition function and derive TD properties

How to account for phonon dispersion?

Folding of phonon bands. Supercell approach
More and more k points can be projected in Γ by increasing the size of the supercell

$$\begin{aligned} a_1 &= a \\ a_2 &= 2a \end{aligned}$$



Phonon Dispersion

SCELLPHONO

Supercell for phonon dispersion

SILICON BULK –
CRYSTAL

0 0 0

.....

SCELPHONO

2 0 0

0 2 0

0 0 2

FREQCALC

DISPERSI

END

A supercell is constructed, as in the case of keyword SUPERCELL as linear combination of the primitive cell unit vectors.

The atomic ordering of the resulting supercell is different with respect to the SUPERCELL case. The first atoms in the list are the ones of the primitive cell, as required for phonon dispersion calculations.

to run a phonon dispersion calculation
rather than a Gamma-only one

Phonon Dispersion: Pband & Pdos

SILICON BULK –
CRYSTAL

SCELPHONO

2 0 0

0 2 0

0 0 2

FREQCALC

RESTART

DISPERSI

BANDS

.....

END

path

[Bilbao Crystallographic Server](http://www.cryst.ehu.es)
<http://www.cryst.ehu.es>

SCELPHONO

2 0 0

0 2 0

0 0 2

Warning !!!

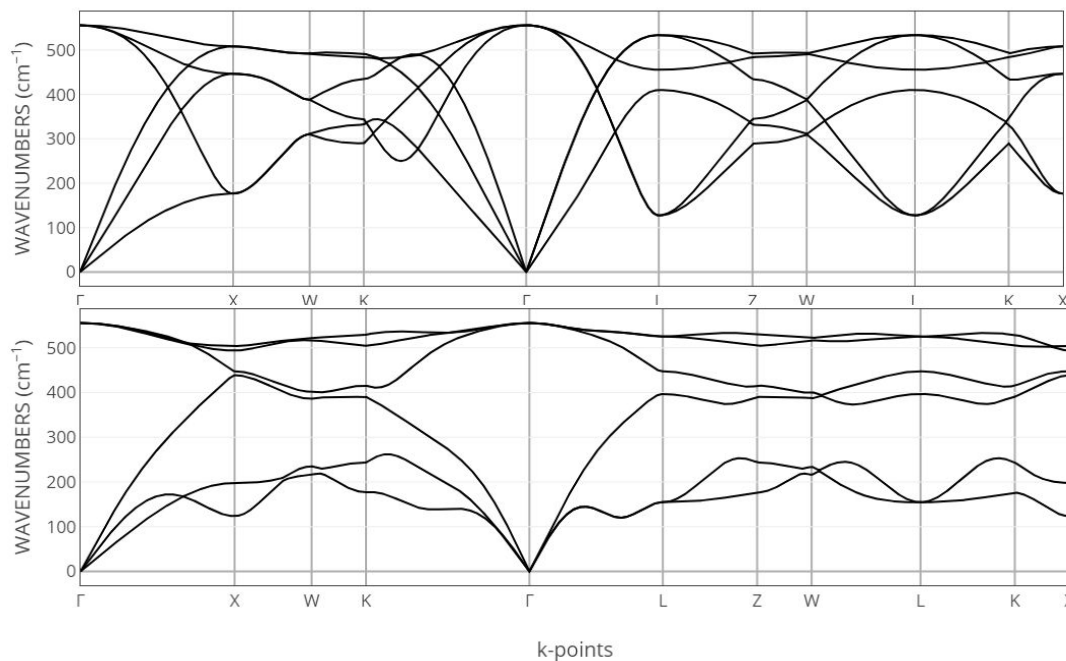
SCELPHONO

3 0 0

0 3 0

0 0 3

PHONON BAND STRUCTURE



SILICON BULK –

SCELPHONO

3 0 0

0 3 0

0 0 3

FREQCALC

RESTART

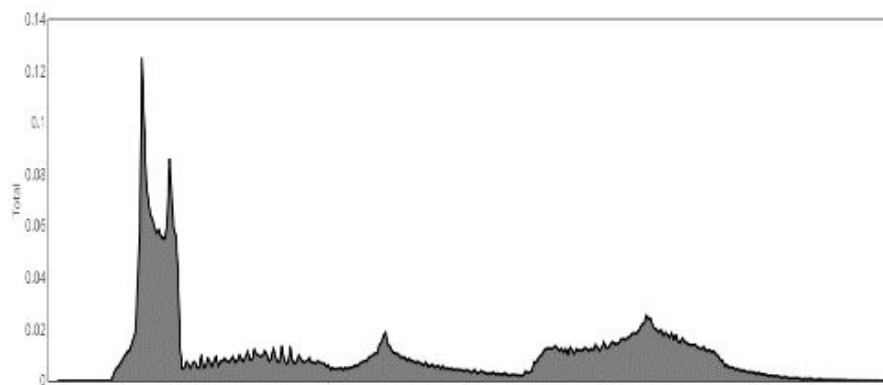
DISPERSI

PDOS

---- frequency range

ENDFREQ

END



Vibrational partition function

```
MgO .....  
CRYSTAL  
0 0 0  
225  
4.20342768  
2  
12 0.0 0.0 0.0  
8 0.5 0.5 0.5  
SCELPHONO  
4. 0. 0.  
0. 4. 0.  
0. 0. 4.  
FREQCALC  
DISPERSI  
.....  
ENDG
```

$$Q_{\text{vib}}(T) = \prod_{s,k} \frac{\exp\left(-\frac{\hbar\omega(s,k)}{2k_B T}\right)}{1 - \exp\left(-\frac{\hbar\omega(s,k)}{2k_B T}\right)}$$



$$G(T) = E + pV + RT\ln Q$$

$$S(T) = RT\ln Q + RT \left(\frac{\partial \ln Q}{\partial T} \right)_V$$

$$C_V(T) = RT\ln Q + RT \left(\frac{\partial^2 \ln Q}{\partial (1/T)^2} \right)_V$$

Accuracy of TD computed variables depends of k sampling

TD variables printed at the end of any FREQ calculations:

without **SCELPHONO/ DISPERSI** computed only in Gamma

THERMODYNAMICAL PROPERTIES ARE CALCULATED AS A SUM OVER 381 MODES

HARMONIC VIBRATIONAL CONTRIBUTIONS TO THERMODYNAMIC FUNCTIONS AT GIVEN
TEMPERATURE AND PRESSURE:

(EL = ELECTRONIC ENERGY
E0 = ZERO-POINT ENERGY
ET = THERMAL CONTRIBUTION TO THE VIBRATIONAL ENERGY
PV = PRESSURE * VOLUME
TS = TEMPERATURE * ENTROPY)

		AU/CELL	EV/CELL	KJ/MOL
EL	:	-275.333017701750	-7492.192307361306	-722886.73610273
E0	:	0.000063808089	0.001736306382	0.16752811

THERMODYNAMIC FUNCTIONS WITH VIBRATIONAL CONTRIBUTIONS

AT (T = 173.00 K, P = 0.10132500E+00 MPA):

		AU/CELL	EV/CELL	KJ/MOL
ET	:	0.000889220636	0.024196923650	2.33464845
PV	:	0.000000431525	0.000011742404	0.00113297
TS	:	0.004479174037	0.121884522046	11.76006978
ET+PV-TS	:	-0.003589521876	-0.097675855993	-9.42428836
EL+E0+ET+PV-TS:		-275.336543415537	-7492.288246910916	-722895.99286297

OTHER THERMODYNAMIC FUNCTIONS:

		<u>mHARTREE/ (CELL*K)</u>	<u>mEV/ (CELL*K)</u>	J/ (MOL*K)
ENTROPY	:	0.025891179407	0.704534809516	67.97728195
HEAT CAPACITY	:	0.005476105508	0.149012406510	14.37751298

TT END TELAPSE 1.29 TCPU 1.29

Thank you all for your kind attention

CRYSTAL

- Features
- Theoretical background
- How to cite
- CRYSTAL User's Manual

CRYSPLIT

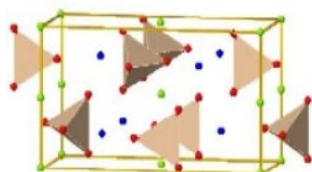
SOFTWARE

- How to get a copy
- License fee
- Platforms

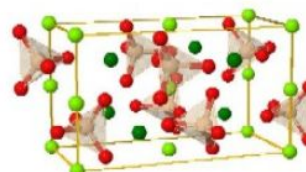
DOCS & SETS

- Documentation
- Encoded Packages
 - CRYSCOR (only CRYSTAL14)
 - TOPOND
- Tutorials
- Basis Sets
- Applications
- Animations of vibrational modes

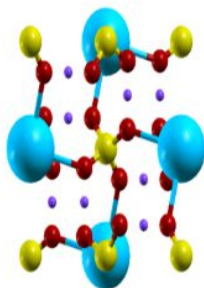
By combining MOLDRW features and a series of Python scripts (written by Yves Noel) the full set of frequencies can be rendered on the web by means of the powerful JMOL engine.
In the following a few examples are provided, taken from published and unpublished works.



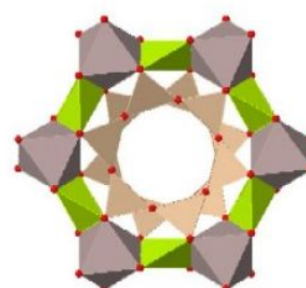
Monticellite



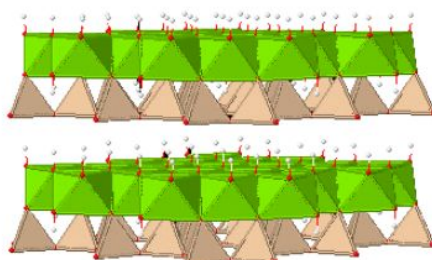
Forsterite



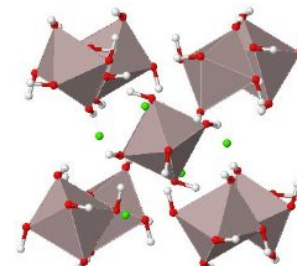
KDP



Beryl



Lizardite



Katoite

<http://www.crystal.unito.it/animations-of-vibrational-modes.php>